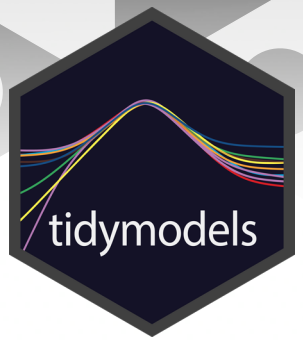


Machine learning with tidymodels : : CHEAT SHEET



Intro

tidymodels is a collection of R packages for modeling and machine learning that share a common design and grammar. Unlike most cheat sheets, which cover the functions of a single package, *this cheatsheet maps the packages themselves*, grouping each by where it fits in the machine learning workflow.

Resampling

Split and resample data for honest evaluation



rsample - Create and manage resampling sets such as cross-validation, bootstraps, and validation splits.



spatialsample - Resample spatial data while respecting geographic structure.

Pre-processing

Prepare data for modeling



recipes - The preprocessing framework for building pipeable feature-engineering steps.



embed - Recipe steps that encode categorical predictors via target, likelihood, and entity embeddings.



textrecipes - Recipe steps that turn text into model-ready features.

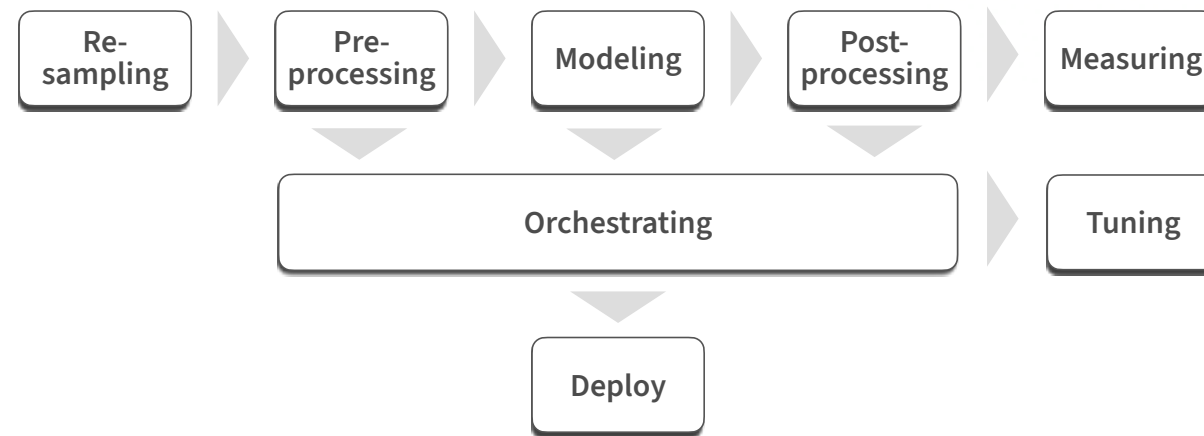


themis - Recipe steps to rebalance class-imbalanced data with up-sampling, down-sampling, and SMOTE.



filtro - Filter-based supervised feature selection.

Machine Learning workflow



Modeling

Define and fit models through one consistent interface



parsnip - The core tidy interface every model plugs into, across many engines.

SPECIALIZED PROBLEMS



poissonreg - Poisson and count-data regression.



censored - Survival models for time-to-event outcomes.



multilevelmod - Mixed-effects and hierarchical models.



tidyclust - Clustering models under a tidy interface



tabby - For tabular deep-learning models, runs with brulee and tabpfn

CLASSIFICATION & REGRESSION



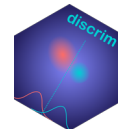
bonsai - Tree-based engines such as LightGBM and partykit.



baguette - Bagged ensembles of trees and MARS.



rules - Rule-based models such as Cubist, C5.0, and RuleFit.



discrim - Discriminant analysis and naive Bayes models.



plsmod - Partial least squares and other projection models.



agua - Interface to h2o models and AutoML.

Post-processing

Adjust predictions



probably - Tune classification thresholds and handle equivocal zones.

tailor - Post-process predictions through calibration and other sequential adjustments.

Measuring

Measure model quality



yardstick - Measure model performance with tidy metrics.



tidyposterior - Compare models across resamples using Bayesian methods.

Orchestrating

Tie the pieces together



workflows - Bundle preprocessing, model, and post-processing into one object.

workflowsets - Create and evaluate many workflows at once.



stacks - Build stacked ensembles from tuned models.

Machine learning with tidymodels :: CHEAT SHEET

tidymodels

Tuning

Optimize hyperparameters



tune - Run grid and iterative hyperparameter search.



dials - Define tuning parameters and build grids. It is often called for you by tune.

METHODS & OPTIMIZATION



finetune - Add search strategies such as racing and simulated annealing.



important - Measure predictor importance.

Deploy

Put models into production

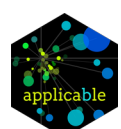
PREPARE & SERVE



vetiver - Version, deploy, and monitor models in production.



butcher - Strip fitted models down to reduce object size.



applicable - Flag new samples that fall outside the training distribution.

RUN IN A DATABASE



tidypredict - Generate SQL to score models inside a database.



modeldb - Fit models directly in a database.



orbital - Convert workflows into portable equations that can run in database.



Data

Datasets used in documentation, tests, and teaching

modeldata - Over 40 example datasets bundled for modeling.

modeldatatoo - Over half a dozen larger datasets downloaded on demand.

Deep Learning

R packages that implement or wrap tabular deep learning models



brulee - Torch-based models, from MLPs to ResNet and SAINT.



tabpfn - A pretrained transformer that predicts tabular data with no training.

Other

GENERAL



broom - Convert model objects into tidy tibbles.



infer - Run statistical inference and hypothesis tests.



corr - Explore correlations in a data frame.

DEVELOPMENT



hardhat - Scaffold new modeling packages.



The **tidymodels** package installs, and loads a set of packages that are considered important during day-to-day machine learning development. It loads the following packages from tidymodels:

rsample	dials
recipes	workflows
parsnip	workflowsets
yardstick	broom
tailor	infer
tune	modeldata

It loads the following packages from the tidyverse:

dplyr	purrr
ggplot2	tidyr

Example

```
library(tidymodels)
```

```
# Split data and make CV folds
set.seed(857)
```

```
splits <- ames |>
  initial_split(prop = 0.8)
```

```
train <- training(splits)
```

```
folds <- train |>
  vfold_cv(v = 5)
```

```
# Feature engineering in a recipe
```

```
rec <- recipe(
  Sale_Price ~ Gr_Liv_Area +
    Year_Built + Bldg_Type,
  data = train
) |>
step_log(Gr_Liv_Area, base = 10) |>
step_dummy(
  all_nominal_predictors()
) |>
step_normalize(
  all_numeric_predictors()
)
```

Example (cont')

```
# Model with parameters to tune
mod <- decision_tree(
  cost_complexity = tune(),
  tree_depth = tune()
) |>
set_engine("rpart") |>
set_mode("regression")
```

```
# Bundle into a workflow
wf <- workflow() |>
  add_recipe(rec) |>
  add_model(mod)
```

```
# Tune over the folds
res <- wf |>
  tune_grid(
    resamples = folds,
    grid = 10
  )
```

```
# Finalize best, refit, test once
best <- res |>
  select_best(
    metric = "rmse"
  )
```

```
final <- wf |>
  finalize_workflow(best) |>
  last_fit(splits)
```

```
collect_metrics(final)
```

```
# Deploy the fitted workflow
library(vetiver)
library(pins)
```

```
v <- final |>
  extract_workflow() |>
  vetiver_model("ames_tree")
```

```
board <- board_temp()
```

```
vetiver_pin_write(board, v)
```